

HEL-Net, Explained

How a team of different neural networks finds four kinds of eye-disease lesions in retinal photographs — told from zero

Based on: Wu, Xia, Song, Lan, “HEL-Net: Heterogeneous Ensemble Learning for comprehensive diabetic retinopathy multi-lesion segmentation via Mamba-UNet”,

Image and Vision Computing 166 (2026) 105879, and the accompanying “HEL-Net, Decoded” analysis.

What you will be able to do after reading

1. Explain **what problem** the paper solves and **why it is hard**, in plain words.
2. Walk through the HEL-Net pipeline **one step at a time**: data augmentation → coarse map → data fusion → specialist branches → feature aggregation → loss.
3. Understand the key maths (Poisson blending, state-space models / Mamba, Dice and IoU) through **small worked numerical examples**.
4. Read the results tables like a reviewer, spot the **weak points**, and reflect on the **AI-ethics questions** that medical-AI papers raise.

How to use this guide

Every step follows the same rhythm: **Big idea** (navy) → **Everyday analogy** (gold) → the technical detail → a **Worked example** (teal) → a **Step recap**. Red boxes flag mistakes or ambiguities in the paper; purple boxes are **AI ethics** reflections. A self-check quiz with answers is at the end.

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The paper in one minute

The whole paper in four sentences

1. **Problem.** Diabetic retinopathy (DR) damages the retina and leaves four kinds of tiny marks (“lesions”) that doctors look for; finding all four automatically, pixel by pixel, is hard because they differ hugely in size, shape and position, and they are very rare.
2. **Idea.** Instead of one network that must be good at everything, use a *team of different specialists*: a first network draws a rough map, then four specialist networks — each tuned to one lesion type — refine it.
3. **Twist.** The specialists are *heterogeneous*: two are classic U-Nets (good at small local detail) and two are Mamba-UNets (good at long-range, global context). A tiny learned “voting” layer combines their answers.
4. **Evidence.** On two public datasets (IDRiD and DDR) HEL-Net beats 14 earlier methods on most metrics, e.g. mean Dice 67.40%, mean IoU 51.99%, mean AUPR 69.52% on IDRiD.

A hospital analogy

Think of a junior doctor who first glances at the photograph and circles “suspicious areas” (**Stage 1**). The photo plus these circles is then sent to four consultants — one per lesion type (**Stage 2**). Some consultants are detail-oriented (U-Net), others look at the whole picture (Mamba-UNet). Finally a chairperson decides, lesion by lesion, *whose opinion to trust most* (**Feature Aggregation Block**).

The paper’s anatomy

Section	What it does	Pages	Figs / Tables
Abstract + 1 Introduction	Problem, gap, proposal, contributions	1.6	1 / –
2 Related work	CNN / Transformer / Mamba families	1.2	–
3 Methodology	Augmentation, HEL-Net, Mamba-UNet, loss	3.5	4 / 1
4 Experiments	Data, comparisons, statistics, ablations, cost	6.2	6 / 10
5 Conclusion	Summary, limitation, future work	0.6	–

Notice that almost half the paper is *evidence*. In applied medical imaging, the experiments section is what convinces reviewers.

Background you need first

This section builds the vocabulary. If you already know a concept, skip its paragraph.

Diabetic retinopathy and its four lesions

A fundus photograph is a colour picture of the back of the eye (the retina). In DR, damaged blood vessels create four lesion types:

Code	Name	What it looks like	Typical size
MA	Microaneurysm	tiny red dots — small bulges in capillaries	a few pixels
HE	Haemorrhage	dark-red blots of leaked blood	small to large
EX	Hard exudate	bright yellow, sharp-edged deposits of fat/protein	small to large clusters
SE	Soft exudate (cotton-wool spot)	pale, fluffy white patches	medium

What is “segmentation”?

Classification vs. segmentation

Classification answers “Does this image contain DR?” (one label per image). *Segmentation* answers “**Which pixels** belong to each lesion?” (one label per pixel). Multi-lesion segmentation outputs **four probability maps** — one per lesion type — each the same size as the image.

The three model families

CNN / U-Net (2015)

A convolutional network shaped like a “U”: an *encoder* shrinks the image and extracts features; a *decoder* grows it back to full size; *skip connections* copy fine detail across. Strength: sharp local detail. Weakness: each filter only “sees” a small window, so understanding the whole retina at once is hard.

Transformer

Uses *self-attention*: every image patch compares itself with every other patch. Strength: global context. Weakness: cost grows with the **square** of the number of patches, $\mathcal{O}(L^2)$.

Mamba / State-Space Model (SSM)

Reads the patches as a sequence and carries a running “memory” h_t forward, like reading a book and keeping notes. Strength: long-range context at **linear** cost $\mathcal{O}(L)$. *Mamba-UNet* is a U-Net whose building blocks are Mamba blocks.

Why linear vs. quadratic matters

HEL-Net resizes images to 800×800 and cuts them into 4×4 patches:

$$L = \frac{800}{4} \times \frac{800}{4} = 200 \times 200 = 40\,000 \text{ tokens.}$$

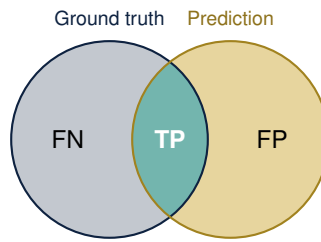
Self-attention would compute $L^2 = 1.6 \times 10^9$ pairwise scores per head per layer. A Mamba scan does work proportional to $L = 4 \times 10^4$ — roughly **40 000 times less**. That is why Mamba is attractive for high-resolution medical images.

What is an ensemble, and what makes it “heterogeneous”?

An *ensemble* combines several models’ predictions, like asking several experts and voting. *Homogeneous* = all experts have the same architecture. *Heterogeneous* = experts are **different kinds** of networks, so they make *different* mistakes — and different mistakes cancel better when combined.

How segmentation is scored

Compare the predicted pixels with the doctor-annotated “ground truth”. Count **TP** (lesion pixels found), **FP** (false alarms) and **FN** (lesion pixels missed).



$$\text{Dice} = \frac{2TP}{2TP + FP + FN}, \quad \text{IoU} = \frac{TP}{TP + FP + FN}, \quad \text{IoU} = \frac{\text{Dice}}{2 - \text{Dice}}.$$

AUPR (area under the precision–recall curve) summarises performance over *all* thresholds, so it does not depend on choosing a cut-off such as 0.5. The prefix “m” (mDice, mIoU, mAUPR) means the average over the four lesion types.

Dice and IoU on a toy image

Suppose for one lesion $TP = 80$, $FP = 10$, $FN = 20$ pixels.

$$\text{Dice} = \frac{160}{160 + 10 + 20} = \frac{160}{190} = 0.842, \quad \text{IoU} = \frac{80}{80 + 10 + 20} = \frac{80}{110} = 0.727.$$

Check the identity: $0.842 / (2 - 0.842) = 0.842 / 1.158 = 0.727$ ✓. **Lesson:** Dice and IoU always rank models the same way, so the paper’s AUPR column is the one that adds genuinely independent evidence.

Step recap

Four lesions (MA, HE, EX, SE) · per-pixel labelling · U-Net = local, Transformer = global but expensive, Mamba = global and cheap · heterogeneous ensemble = different experts · Dice / IoU / AUPR measure overlap.

Step 0 — Why is this problem hard?

The paper names **two challenges**:

1. **Extreme diversity.** An MA may be 3–5 pixels; an exudate cluster may cover a large region. One network rarely handles both scales well.
2. **Local vs. global conflict.** Small lesions need fine local detail (CNN strength); big, scattered lesions need long-range context (Transformer/Mamba strength). Most models choose one.

A third, hidden challenge is **class imbalance**:

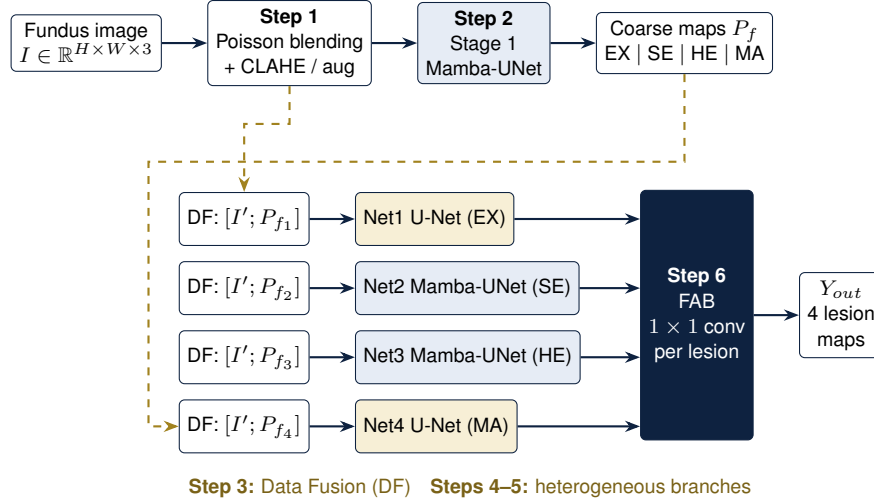
Pixel type (IDRiD)	Share of all pixels
Background (no lesion)	97.80%
All lesions together	≈ 2.2%
Soft exudates (SE)	0.19%
Microaneurysms (MA)	0.10%

How rare is 0.10%?

An 800×800 image has 640 000 pixels. 0.10% of that is about **640 MA pixels** in the whole image. A lazy network that predicts “no MA anywhere” is already 99.9% accurate — which is why plain accuracy is useless here and why Dice/IoU/AUPR and weighted losses are used.

The datasets are also small: **IDRiD** has 81 images (54 train / 27 test) and **DDR** has 757 images (383 train / 149 validation / 225 test).

The full pipeline at a glance



Blue fill = Mamba-UNet, gold fill = U-Net. Dashed arrows carry the augmented image I' and the coarse maps into the Data Fusion blocks. Each branch predicts all four classes; the FAB fuses them class by class.

We now walk through each step.

Step 1 — Make more training data with Poisson blending

Big idea

There are too few lesion examples, especially MAs. So the authors **cut lesions out of training images and paste them into other training images** — but seamlessly, so the network cannot tell they were pasted.

Photoshop’s “healing brush”

If you paste a sticker on a photo, the edge is obvious. A healing brush instead copies the *texture changes* (edges, shading) of the source but matches the *colour at the border* to the destination. Poisson blending is the mathematics behind that brush.

The three sub-steps

1. **Build a lesion library** L — crop EX, SE, HE and MA patches (with their masks) from *training* images only (Eq. 1).
2. **Apply anatomical rules** (Eq. 2): never place a lesion on the optic disc; MAs must also avoid blood vessels (vessel masks come from a vessel-segmentation network).
3. **Blend seamlessly** (Eqs. 3–7) and insert between 0 and β lesions per image.

The maths, gently

Let f be the unknown pixel values inside the paste region Ω , f^* the host image, and g the lesion patch. Poisson blending solves

$$\min_f \iint_{\Omega} |\nabla f - \nabla g|^2 \quad \text{subject to} \quad f = f^* \text{ on the border } \partial\Omega.$$

In words: *inside* the region, make the **gradients** (changes between neighbouring pixels) look like the lesion's; *on the border*, match the host's **colour**. Setting the derivative to zero gives the Poisson equation $\Delta f = \Delta g$ inside Ω — a large but sparse linear system. In practice OpenCV's `cv2.seamlessClone` does this in one call.

A 1-D Poisson blend you can do by hand

Host row: $f^* = [10, 10, 10, 10, 10]$ (a darker retina). Source row containing the lesion: $g = [50, 50, 60, 50, 50]$ (a bright bump of height 10 on a lighter background). We paste the three middle pixels.

Naïve paste: $[10, 50, 60, 50, 10]$ — jumps of +40 and −40 at the edges (a visible seam).

Poisson paste: keep the two border pixels at 10 and, for each inner pixel $i = 1, 2, 3$, solve

$$2f_i - f_{i-1} - f_{i+1} = 2g_i - g_{i-1} - g_{i+1} \quad (\text{discrete } \Delta f = \Delta g).$$

The right-hand sides are $(-10, 20, -10)$. The first and third equations give $f_1 = f_3 = f_2/2$; substituting in the middle one gives $f_2 = 20$, so

$$f = [10, 10, 20, 10, 10].$$

The *shape* of the lesion (a bump of height 10) is preserved exactly, but it now sits on the host's brightness level — no seam.

Careful reader's note

- The maximum number of inserted lesions β is never given, so the step cannot be reproduced exactly.
- The abstract reports the *no-augmentation* variant (best mAU_{PR}), while the Poisson variant has better mDice/mIoU (68.58% / 53.46%) and is later labelled “HEL-Net (Ours)” in Table 5. A paper should use one headline variant throughout.
- This idea is adapted from earlier work (PBDA, Pérez et al.); the authors say so honestly.

Data leakage

Building the lesion library from *test* images would leak test information into training and inflate results. The library must come from training images only — a good habit to state explicitly in any paper that uses copy-paste augmentation.

Step recap

Lesion library → anatomical placement rules → seamless Poisson paste. Purpose: more and more varied examples of rare lesions.

Step 2 — Stage 1: draw a rough map

Big idea

The enhanced image I' passes through a **Mamba-UNet** which outputs $P_f \in \mathbb{R}^{H \times W \times 4}$: four rough probability maps, one per lesion (Eq. 8). Slice k , written $P_{f_k} = P_f[:, :, k]$, is a *prior* that says “lesion k is probably here”.

Everyday analogy

A highlighter pass over a textbook: not precise, but it tells the careful reader where to focus.

Why Mamba-UNet here? Stage 1 needs to see the *whole* retina to decide roughly where lesions are — a global task. The ablation study (Step 9) shows this choice is **load-bearing**: when Stage 1 is a plain U-Net, every two-stage variant performs *worse* than a single U-Net. A bad rough map misleads every specialist after it.

Step recap

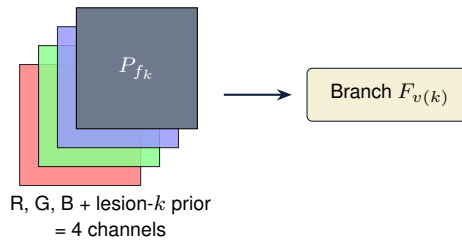
Stage 1 = a global “where to look” map, produced by Mamba-UNet. Its quality decides everything downstream.

Step 3 — Data Fusion: give each specialist a hint**Big idea**

For branch k , stack the 3-channel colour image with the 1-channel prior of lesion k :

$$x_{v(k)} = [I'; P_{f_k}] \in \mathbb{R}^{H \times W \times 4} \quad (\text{Eq. 9}).$$

That is all “Data Fusion” is: a **fourth input channel**.



The paper brands this the “*multi-perspective lesion navigation strategy*”. Giving a simple operation a memorable name is a common (and legitimate) writing device: it makes the idea easy to cite. A variant called HEL-Net-A, which feeds *all four* priors to every branch, was also tested (Table 8); one prior per branch worked better on IDRiD (about +0.87 mIoU / +0.71 mDice).

Step recap

Each specialist receives the image *plus* a “look here” map for its own lesion.

Step 4 — The heterogeneous specialist branches

Each branch k produces a full four-class prediction $P_{v(k)} = F_{v(k)}(x_{v(k)})$ (Eq. 10). The assignment of backbones is:

Branch	Specialises in	Backbone	Parameters (approx.)
Net1	EX (hard exudates)	U-Net	13.40 M
Net2	SE (soft exudates)	Mamba-UNet	19.12 M
Net3	HE (haemorrhages)	Mamba-UNet	19.12 M
Net4	MA (microaneurysms)	U-Net	13.40 M

Where does specialisation come from? Two signals:

1. the **prior channel** (*where to look*), and
2. the **loss weights**, which double the importance of the branch's own lesion (*what to care about*; see Step 7).

Counting HEL-Net's parameters

Stage 1 (Mamba-UNet) + two Mamba-UNet branches + two U-Net branches:

$$19.12 + 2(19.12) + 2(13.40) = 19.12 + 38.24 + 26.80 = \mathbf{84.16\ M}.$$

This matches Table 7 ✓. (The FAB adds only ≈ 20 parameters — see Step 5.)

Rationale vs. design mismatch

The introduction says U-Net is used for *small* lesions, yet U-Net is assigned to EX, which are among the *largest* lesions. The ablation supports the assignment empirically, but the written reason does not match it. Lesson for your own writing: let the rationale follow the evidence.

Step recap

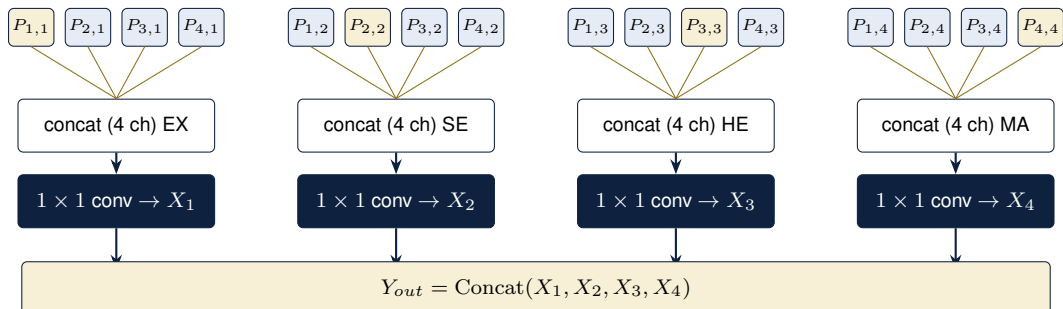
Four specialists, two architectures. Specialisation = prior channel + loss emphasis.

Step 5 — Feature Aggregation Block: the learned vote

Big idea

Each of the 4 branches outputs 4 class maps $\Rightarrow$ 16 maps. For lesion c , the FAB takes the four class- c maps (one from each branch), stacks them, and mixes them with a 1×1 convolution (Eq. 11):

$$X_c = w_{c,1}P_{1,c} + w_{c,2}P_{2,c} + w_{c,3}P_{3,c} + w_{c,4}P_{4,c} + b_c, \quad Y_{out} = \text{Concat}(X_1, \dots, X_4).$$



Weighted voting

For haemorrhages, the chairperson might learn: “trust the HE specialist 60%, the SE specialist 20%, the others 10% each.” For microaneurysms the weights would be different. The FAB *learns* these trust levels from data.

FAB on one pixel

Suppose at one pixel the four branches give HE probabilities 0.9, 0.4, 0.8, 0.2 and the learned weights are $w = (0.1, 0.2, 0.6, 0.1)$, $b = 0$:

$$X_{\text{HE}} = 0.1(0.9) + 0.2(0.4) + 0.6(0.8) + 0.1(0.2) = 0.09 + 0.08 + 0.48 + 0.02 = 0.67.$$

Size of the FAB: 4 weights + 1 bias = 5 per class, $\times 4$ classes = **20** parameters. Tiny, yet the ablation credits it with +1.44 mAUPR on IDRiD.

Careful reader’s note

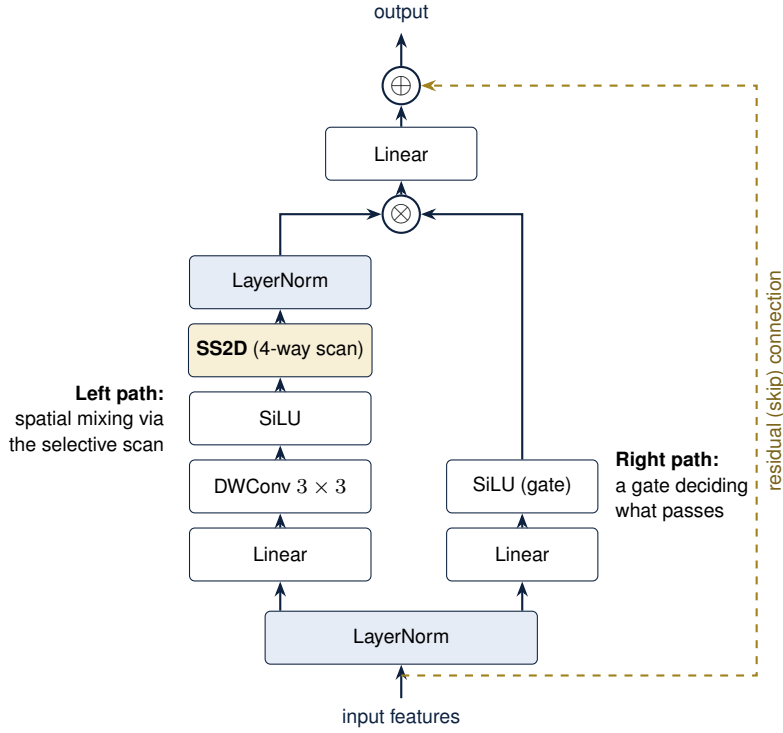
The text claims the FAB improves “all four lesion types”. True on IDRiD, but on DDR the SE score *drops* ($41.27 \rightarrow 39.63$), and the reported DDR mAUPR gain (+1.66) recomputes to $46.22 - 45.07 = +1.15$.

Step 6 — Inside Mamba-UNet and the VSS block**The architecture**

Stage	Resolution	Channels
Patch embedding (4×4)	$H/4 \times W/4$	C
Encoder 1–4 (2 VSS blocks each + patch merging)	$1/4, 1/8, 1/16, 1/32$	$C, 2C, 4C, 8C$
Decoder (2 VSS blocks each + patch expanding + skips)	back to $1/4$	$4C, 2C, C$
Linear projection	$H \times W$	number of classes

It is a U-Net in which every convolutional block is replaced by a **Visual State Space (VSS)** block.

The VSS block



SS2D (2-D selective scan) is the heart. An image is 2-D but Mamba reads 1-D sequences, so the feature map is unrolled in **four directions** (left→right, right→left, top→bottom, bottom→top), each sequence is scanned by a selective SSM, and the four results are merged. This removes the bias of reading the image in only one direction.

The state-space model — step by step

(a) **Continuous form (Eq. 12).** A hidden state $h(t)$ evolves with the input $x(t)$:

$$h'(t) = A h(t) + B x(t), \quad y(t) = C h(t).$$

(b) **Discretise (Eq. 13).** Computers work in steps of size Δ . Holding x constant during a step (“zero-order hold”):

$$\bar{A} = e^{\Delta A}, \quad \bar{B} = (\Delta A)^{-1}(e^{\Delta A} - I) \Delta B, \quad h_t = \bar{A} h_{t-1} + \bar{B} x_t, \quad y_t = C h_t.$$

(c) **Selective = input-dependent.** Mamba makes B , C and Δ *functions of the input*, so the model can decide, token by token, what to remember and what to forget. It is still computed by a parallel scan with cost $\mathcal{O}(L)$.

A 1-D SSM with three numbers

Take a scalar state with $A = -1$, $B = 1$, $C = 1$, step $\Delta = 0.5$.

$$\bar{A} = e^{-0.5} = 0.607, \quad \bar{B} = \frac{e^{-0.5} - 1}{-0.5} \times 0.5 = 1 - e^{-0.5} = 0.393.$$

Feed the input sequence $x = [1, 0, 0]$ (a single “spike”), starting with $h_0 = 0$:

$$h_1 = 0.393, \quad h_2 = 0.607 \times 0.393 = 0.239, \quad h_3 = 0.607 \times 0.239 = 0.145.$$

The state **remembers** the spike and **fades** it gradually. A larger Δ would forget faster; a smaller Δ would remember longer. *Selective* Mamba learns to choose Δ per token — e.g. remember lesion-like patches longer, forget plain background quickly.

Notation slips in the paper

Eq. 13 is printed as $y_t = Cx_t$; it should be $y_t = Ch_t$. The convolution kernel “ K ” is mentioned but never defined (it is $\bar{K} = (C\bar{B}, C\bar{A}\bar{B}, \dots, C\bar{A}^{L-1}\bar{B})$). The symbol C is also reused for RGB channels, number of classes and the SSM output matrix.

Step recap

Mamba-UNet = U-Net built from VSS blocks. VSS = gated block around SS2D. SS2D = four-direction selective scan. The SSM keeps a fading memory whose fading rate is learned per token.

Step 7 — Training with triple supervision**Big idea**

The network is corrected at **three points** (Eq. 14):

$$\mathcal{L}_{seg} = \underbrace{\mathcal{L}_M}_{\text{stage 1}} + \underbrace{\sum_{k=1}^4 \mathcal{L}_S^k}_{\text{each branch}} + \underbrace{\mathcal{L}_O}_{\text{fused output}}.$$

$\mathcal{L}_M$ and $\mathcal{L}_S^k$ are weighted cross-entropy; $\mathcal{L}_O$ is weighted binary cross-entropy.

Everyday analogy

A teacher who checks the rough draft, each chapter, and the final essay — rather than only the final essay — gives feedback that reaches every part of the work.

The weight table (Table 1)

Weight	Applies to	EX	SE	HE	MA	BG
w^M	stage 1	1	2	1	2	0.1
$w_{c,1}^S$	branch EX	2	2	1	2	0.1
$w_{c,2}^S$	branch SE	1	4	1	2	0.1
$w_{c,3}^S$	branch HE	1	2	2	2	0.1
$w_{c,4}^S$	branch MA	1	2	1	4	0.1
w_O^S	fused output	10	40	10	40	—

How to read it. Start from base weights (EX 1, SE 2, HE 1, MA 2, background 0.1). The rare classes SE and MA get twice the weight. For branch k , multiply *its own* lesion’s weight by $T = 2$ (bold entries). Background gets 0.1 so the 97.8% of empty pixels do not dominate.

Why weight the loss?

Imagine 1000 background pixels and 1 MA pixel, each with error 1. Unweighted loss: background contributes 1000, MA contributes 1 — the network ignores MA. With weights 0.1 and 4: background = 100, MA = 4. Still small, but 40× more influential than before. Weights shift attention toward what matters clinically.

Training schedule is ambiguous

The paper says the branches are “trained independently” and that the ensemble is “jointly optimised”. Whether Stage 1 is frozen, detached, or trained end-to-end with the branches is never stated. A short algorithm box would remove the ambiguity.

Training settings

Pre-processing	CLAHE (clip 2, grid 8), resize to 800×800
Augmentation	rotation $\pm 20^\circ$, flips, colour jitter, Poisson blending
Optimiser	Adam, lr 2×10^{-4} , $\beta_1 = 0.5$, $\beta_2 = 0.999$
Schedule	lr $\times 0.9$ every 100 epochs, 400 epochs, batch size 2
Hardware	PyTorch, one RTX A6000 (48 GB)

Step 8 — Reading the results

Headline comparison

Dataset	Metric	HEL-Net	Strongest other	Δ (points)
IDRiD	mIoU	51.99	49.94 (M2MRF)	+2.05
IDRiD	mDice	67.40	65.71 (M2MRF)	+1.69
IDRiD	mAUPR	69.52	68.79 (HACDR-Net)	+0.73
DDR	mIoU	39.05	33.70 (HACDR-Net)	+5.35
DDR	mDice	55.25	49.30 (HACDR-Net)	+5.95
DDR	mAUPR	46.22	50.36 (HACDR-Net)	-4.14

Verify a mean yourself

IDRiD per-lesion Dice for HEL-Net: EX 80.85, SE 73.44, HE 66.32, MA 48.98.

$$\text{mDice} = \frac{80.85 + 73.44 + 66.32 + 48.98}{4} = \frac{269.59}{4} = 67.40$$

Also EX: $\text{IoU} = 0.8085 / (2 - 0.8085) = 67.86\%$, exactly the reported EX IoU ✓.

Per-lesion ranking on IDRiD

Metric	EX	SE	HE	MA
IoU	✓ 1st	✓ 1st	2nd	✓ 1st
Dice	✓ 1st	✓ 1st	2nd	✓ 1st
AUPR	✓ 1st	2nd	✓ 1st	✗ 5th

MA — the tiniest lesion — remains the hardest: MA Dice is only 48.98%. Note also the paper’s other evidence: mean $\pm$ SD across images, t -test p -values, box plots, PR curves and qualitative overlays (yellow = correct, red = false positive, green = missed).

What the DDR AUPR drop means

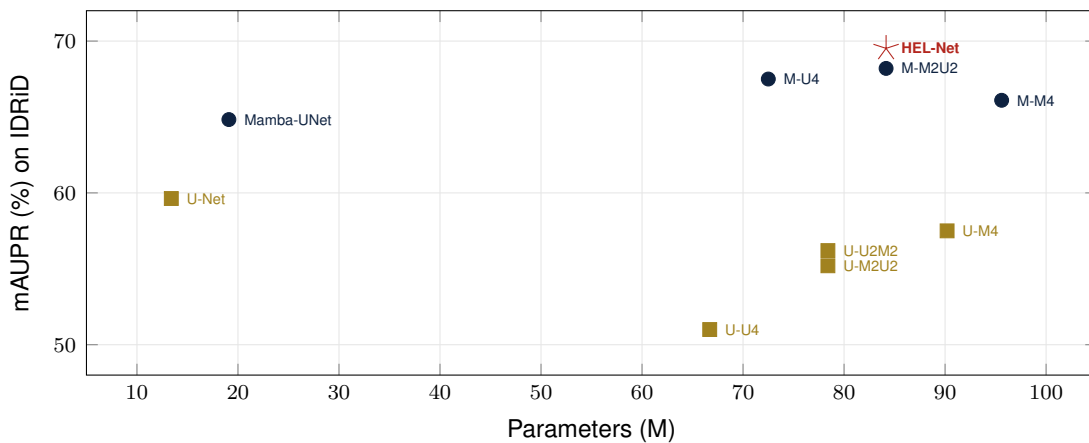
On DDR, HEL-Net is best on Dice/IoU but only about 6th on mAUPR. High Dice with lower AUPR suggests HEL-Net’s advantage comes partly from a well-chosen *threshold* (calibration), not from ranking every pixel better. This is a good example of why reporting a threshold-free metric matters.

Step 9 — Ablations: what each piece contributes

Big idea

An *ablation* removes or swaps one component to see how much it matters. The key question is: **is HEL-Net better because of its design, or just because it is bigger?**

Variants are named by *Stage-1 backbone – branch backbones*: “M” = Mamba-UNet, “U” = U-Net. For example M-U4 means Mamba Stage 1 with four U-Net branches; HEL-Net is effectively M-U2M2.



Navy circles: Mamba-UNet Stage 1. Gold squares: U-Net Stage 1. Red star: HEL-Net. Positions of intermediate variants are approximate, redrawn from the decode.

Three lessons from this plot:

1. **Capacity is not the explanation.** M-M4 has the most parameters but is beaten by smaller variants.
2. **Assignment matters.** HEL-Net and M-M2U2 have *identical* size (84.16 M); they differ only in which lesion gets which backbone — worth +1.32 mAUPR.
3. **A weak prior poisons the cascade.** Every U-Net-first variant falls *below* a plain U-Net. Stage 1 must be strong.

Other ablations: Mamba-UNet beats U-Net alone by 5.20 mAUPR (64.83 – 59.63); the FAB adds +1.44 mAUPR on IDRiD; the branch weight multiplier $T = 2$ was tuned in Table 11.

The cost

HEL-Net needs 484.86 GFLOPs and runs at 6.27 images per second — $1.28\times$ HRNetV2, $1.37\times$ M2MRF and $2.65\times$ HACDR-Net in compute, and the slowest of the compared methods. The paper calls this “slightly higher”; a reader should judge that wording critically.

Step 10 — Thinking like a reviewer

Strengths

- A clear, well-motivated idea: match backbones to lesion characteristics and guide them with priors.
- A rich evidence package: two datasets, SD, p -values, box plots, PR curves, four ablations, complexity analysis.
- The architecture-combination ablation with parameter counts genuinely separates design from size.
- An honest limitation (computational cost) and concrete future work.

Weaknesses

1. **Test-set tuning.** IDRiD has no validation split, so T and thresholds seem chosen on the 27 test images.
2. **Statistics.** The test type is unspecified, $n = 27$, and 42 p -values are reported with no multiple-comparison correction.
3. **Mixed baselines.** Some baseline means are copied from earlier papers while SDs come from re-runs.
4. **Novelty overlap.** Poisson augmentation and Mamba-UNet are adapted; the novelty is the combination.
5. **Reproducibility.** No code, no seeds, patch size and width C unspecified, β missing.
6. **Presentation.** More than ten numeric or labelling inconsistencies (some listed below).

Multiple comparisons in one line

With 42 tests at $\alpha = 0.05$, about $42 \times 0.05 \approx 2$ would look “significant” by pure chance even if nothing were real. The Bonferroni correction uses $\alpha = 0.05/42 \approx 0.0012$. Under it, only 16 of the 39 originally-significant entries survive, and against the two strongest baselines just one metric each.

A sample of the number audit

Where	Claim in text	Recomputed	OK?
§4.4.1 IDRiD	MA AUPR is “second-best”	ranks 5th (45.70 vs 48.80, 46.94, 46.27, 46.01)	✗
§4.4.1 DDR	MA Dice gain “more than 2.26%”	$39.86 - 27.67 = 12.19$	✗
Table 3	Twins-SVT-B mAUPR 63.84	implies a digit swap (63.12 vs 36.12)	✗
§4.5.1	Mamba-UNet beats U-Net by 5.2 mAUPR	$64.83 - 59.63 = 5.20$	✓
Table 7	HEL-Net 84.16 M params	$19.12 + 2(19.12) + 2(13.40)$	✓
§4.5.4	cost “slightly higher”	up to $2.65\times$ GFLOPs, slowest	✗

None of these changes the main conclusion on IDRiD, but each is the kind of detail a careful reviewer uses to question everything else.

Questions a reviewer would ask

1. What does a single network with ≈ 84 M parameters achieve (capacity control)?
2. Is Stage 1 frozen or trained jointly? Is its output detached before Data Fusion?
3. How is the Dice threshold chosen, and on which split?
4. How sensitive are results to resizing ($2848 \times 4288 \rightarrow 800 \times 800$ distorts shape and shrinks MAs by $3.6\text{--}5.4\times$)?
5. Can results be given as mean $\pm$ SD over at least three random seeds?

AI ethics corner — what this paper teaches beyond accuracy

1. Honest evaluation is an ethical duty in medical AI

Tuning hyper-parameters or thresholds on the *test* set makes a model look better than it will be on new patients. In medicine, an over-optimistic number can translate into missed disease. Good practice: train / validation / test splits, report how every choice was made, and keep the test set untouched until the end.

2. Reproducibility and openness

Data “on request” and no released code mean other groups cannot verify the claims. Releasing code, seeds and exact settings is part of scientific accountability — especially when results may influence clinical tools.

3. Statistics and the temptation of selective claims

Reporting 42 *p*-values and highlighting only the favourable ones, without correction, is a mild form of *p-hacking*. The ethical habit is to pre-state which comparisons matter and correct for multiplicity.

4. Generalisation and fairness

IDRiD (81 images) and DDR (757 images) come from specific populations, cameras and clinics. A model trained there may perform worse on other ethnic groups, cameras or image qualities. Before clinical use one needs external validation across diverse populations and subgroup analysis.

5. Honest language about cost and novelty

Calling a $2.65\times$ compute increase “slightly higher”, or claiming “for the first time” without a hedge, nudges readers toward a rosier view. Precise language is part of research integrity. The cost also matters for fairness: heavy models are harder to deploy in low-resource clinics where DR screening is most needed.

6. Synthetic data needs care

Poisson-blended lesions are *synthetic* medical findings. They must never leak into test sets, and their anatomical realism should ideally be checked by clinicians, so the model does not learn shortcuts from artefacts.

Key takeaways

1. **Divide and specialise.** One generalist network struggles with lesions of very different sizes; a team of specialists does better.
2. **Mix local and global.** U-Net (local detail) and Mamba-UNet (long-range, linear cost) complement each other.
3. **Priors guide attention.** A rough map added as a 4th channel tells each specialist where to look — but a weak prior hurts.
4. **Learn the vote.** A 20-parameter 1×1 conv learns which specialist to trust per lesion.
5. **Fight imbalance** with seamless lesion pasting and weighted losses.
6. **Read critically.** Check the arithmetic, the splits, the statistics and the wording — good results deserve careful reporting.

Self-check quiz

1. Why is pixel accuracy a poor metric for MA segmentation?
2. What exactly is “Data Fusion” in HEL-Net?
3. Which lesions are handled by U-Net branches, and which by Mamba-UNet branches?
4. A model has Dice = 0.60 for a lesion. What is its IoU?
5. How many parameters does the FAB have, and why?
6. Why does Mamba scale better than self-attention on an 800×800 image?
7. What does the ablation with U-Net as Stage 1 teach us?
8. In the SSM example, what happens to the memory if Δ becomes larger?
9. Name two reasons the reported IDRiD numbers might be slightly optimistic.
10. Give one ethical concern about deploying HEL-Net in a rural screening programme.

Answers

1. MA covers only $\approx 0.10\%$ of pixels; predicting “no MA” everywhere already gives 99.9% accuracy.
2. Concatenating the 3-channel image with the 1-channel Stage-1 prior for that branch’s lesion (4 channels in total).
3. U-Net: EX and MA. Mamba-UNet: SE and HE (and Stage 1).
4. $\text{IoU} = 0.60 / (2 - 0.60) = 0.60 / 1.40 \approx 0.429$.
5. 20: for each of 4 classes, 4 weights (one per branch) + 1 bias.
6. $L = 40\,000$ tokens; attention costs $\propto L^2 = 1.6 \times 10^9$, Mamba $\propto L$.

7. A weak Stage-1 prior misleads all branches: every U-Net-first cascade is worse than a single U-Net.
8. $\bar{A} = e^{\Delta A}$ gets smaller (with $A < 0$), so the state forgets faster.
9. Hyper-parameters/thresholds chosen on the 27-image test set; no multiple-comparison correction; small n .
10. Any of: heavy compute (≈ 485 GFLOPs, 6 FPS) on low-cost hardware; unproven generalisation to local populations and cameras; no released code for independent validation.

Glossary

Ablation An experiment that removes or swaps one component to measure its contribution.

AUPR Area under the precision–recall curve; threshold-free, suited to rare classes.

CLAHE Contrast-Limited Adaptive Histogram Equalisation: local contrast enhancement.

Cross-entropy (weighted) Classification loss; weights make rare classes count more.

Data Fusion (DF) Stacking the image with a lesion prior as a 4th channel.

Dice / IoU Overlap scores between prediction and ground truth; $\text{IoU} = \text{Dice} / (2 - \text{Dice})$.

Ensemble Combining several models' predictions.

FAB Feature Aggregation Block: class-wise 1×1 convolution that fuses branch outputs.

Heterogeneous Built from different kinds of models (here U-Net and Mamba-UNet).

Mamba A selective state-space model with input-dependent parameters and linear cost.

Poisson blending Seamless pasting that copies gradients inside a region and matches colours on its border.

Prior A rough map that tells a later network where something probably is.

SS2D 2-D selective scan: unroll an image in four directions and scan each with an SSM.

SSM State-space model: $h_t = \bar{A}h_{t-1} + \bar{B}x_t$, $y_t = Ch_t$.

U-Net Encoder–decoder CNN with skip connections, the classic segmentation backbone.

VSS block Visual State Space block: the Mamba building block used inside Mamba-UNet.